K J's EDUCATIONAL INSTITUTES
TRINITY COLLEGE OF ENGINEERING & RESEARCH
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Savitribai Phule Pune University
Accredited NAAC Grade 'A+'
Department of Computer Engineering

DIV:C **Subject: DBMS**

Tourism Management System

Submitted by:

- Prasanna Shinde CO2135
- Veena Shinde – CO2137
- Apeksha Sonawane – CO2143
- Kamlesh Supe – CO2147

Under the Guidance of:

- Prof. Deepika Falak

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1. Abstract

The Smart Tourism Management System is a web-based application developed to simplify and digitize tourism services for Panchgani. The system provides functionalities such as browsing tourist destinations, user registration, login, and hotel booking. It uses a structured MySQL database for efficient data storage and retrieval.

The application follows the Software Development Life Cycle (SDLC) approach, ensuring proper planning, design, implementation, and testing. The frontend is developed using HTML, CSS, and PHP, while the backend uses MySQL. The system improves user experience and reduces manual effort in tourism management.

2. Introduction

Tourism plays a vital role in economic development, especially in hill stations like Panchgani. Traditional tourism management systems are manual and inefficient.

This project aims to:

- Provide an online tourism platform
- Manage tourist data efficiently
- Offer booking services
- Improve accessibility and user experience

3. Software Development Life Cycle (SDLC)

I. Planning:

- Objective: Build a tourism website for Panchgani
- Scope: Tourist info + booking system
- Tools selected: HTML, PHP, MySQL, XAMPP

II. Requirement Analysis:

User Requirements:

- View destinations
- Register/Login
- Book hotels

Admin Requirements:

- Manage users
- Manage bookings
- Update places

III. System Design

Architecture:

- Frontend: HTML, CSS
- Backend: PHP
- Database: MySQL

Database Design:

Entities:

- Tourist
- Booking
- Places

IV. Implementation

- Developed responsive UI (as shown in your screenshot)
- Created database tables
- Connected frontend with database using PHP

V. Testing

- Manual Testing ✓
- Form validation ✓
- Database operations ✓

VI. Deployment

- Deployed on local server using XAMPP
- Access via: http://localhost/panchgani-tourism/index.php

VII. Maintenance

- Add new features (payment gateway, reviews)
- Improve UI/UX
- Security updates

4. Software Requirement Specification (SRS)

◆ Functional Requirements

- User Registration & Login
- View Tourist Places
- Hotel Booking System
- Admin Management

◈ Non-Functional Requirements

- Fast loading
- Secure database
- Responsive design

◈ Hardware Requirements
- System with 4GB RAM minimum

◈ Software Requirements
- XAMPP Server
- MySQL
- Web Browser

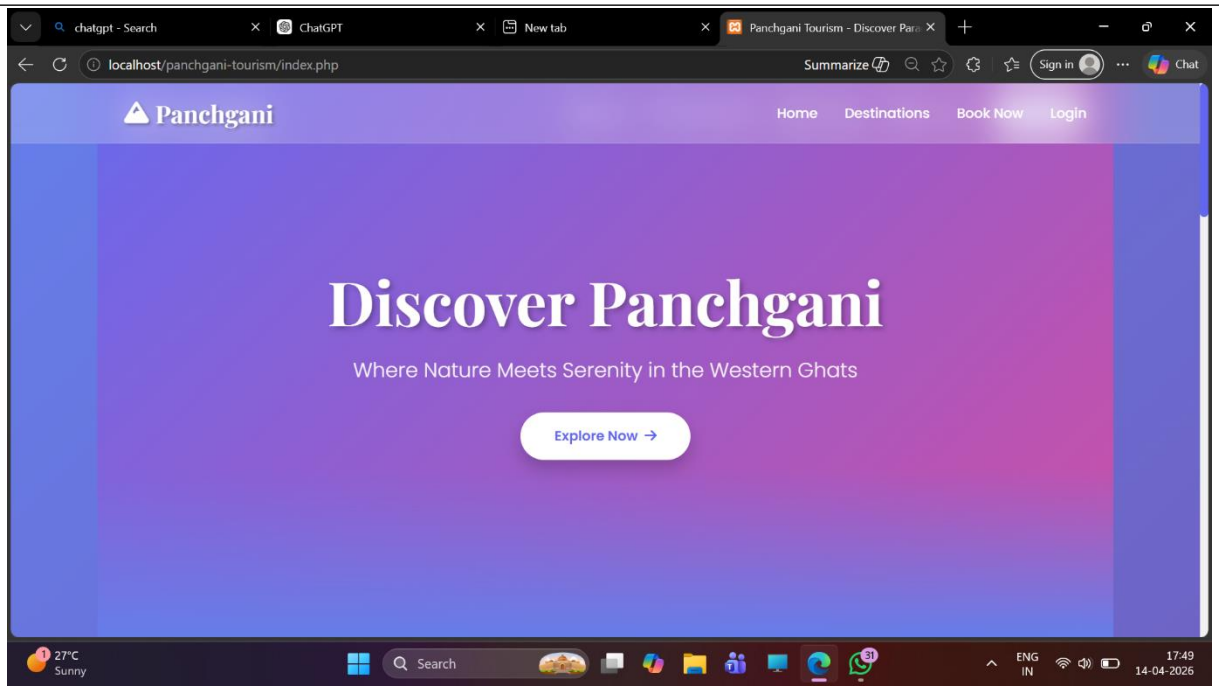
5. DBMS Concepts Used

Entities:
- Tourist (tourist_id, name, email, phone)
- Booking (booking_id, tourist_id, hotel_id, date)
- Place (place_id, name, description)

Relationships:
- Tourist → Books → Place
- Tourist → Visits → Place

6. Graphical User Interface (GUI)

Our website includes:
- Home Page (with gradient hero section)
- Navigation bar (Home, Destinations, Book Now, Login)
- Explore button
- Booking and login forms



Data Connection Code:

```
<?php
session_start();
// Database configuration
define('DB_HOST', 'localhost');
define('DB_USER', 'root');
define('DB_PASS', 'root');
define('DB_NAME', 'panchgani_tourism');
// Create connection
$conn = mysqli_connect(DB_HOST, DB_USER, DB_PASS, DB_NAME);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
// Function to sanitize input
function sanitize($input) {
    global $conn;
```

```
    return mysqli_real_escape_string($conn,
htmlspecialchars(strip_tags(trim($input))));
}
?>
```

7. Testing Document

Manual Testing Table

Test Case	Input	Expected Output	Result
Valid Registration	Correct data	Success message	Pass
Empty Fields	Blank	Error	Pass
Invalid Email	Wrong format	Validation error	Pass

8. Conclusion

The Smart Tourism Management System for Panchgani successfully demonstrates the implementation of database concepts and software engineering principles. The system provides an efficient and user-friendly platform for tourists and administrators.

It reduces manual work, improves data handling, and enhances user experience. The project fulfills all SDLC phases and serves as a strong foundation for future enhancements.